

AHA SCIENTIFIC STATEMENT

Screening, Diagnosis, and Management of Pediatric Hypertriglyceridemia: A Scientific Statement From the American Heart Association

Amy L. Peterson, MD, MS, FAHA, Vice Chair; Ambika P. Ashraf, MD; Jennifer Bachman, MSN, RN, CPNP-PC; Robert A. Hegele, MD, FAHA; Maya Rashad, RD, LD; Andrew M. South, MD, MS; Andrew H. Tran, MD, MPH, MS, FAHA; Justin Zachariah, MD, MPH, FAHA; Jonathan P. Wong, MD, Chair; on behalf of the American Heart Association Cardiovascular Disease Prevention Committee of the Council on Lifelong Congenital Heart Disease and Heart Health in the Young; Council on Clinical Cardiology; Council on Genomic and Precision Medicine; Council on Lifestyle and Cardiometabolic Health; Council on Basic Cardiovascular Sciences; and Council on Cardiovascular and Stroke Nursing

ABSTRACT: Pediatric hypertriglyceridemia is common in youths, affecting 10% to 20% of US children and adolescents, and is increasingly recognized because of its association with obesity, insulin resistance, and steatotic liver disease. Hypertriglyceridemia can lead to life-threatening acute pancreatitis, and prolonged exposure may increase risk of atherosclerotic cardiovascular disease. Screening for pediatric dyslipidemia has been recommended by a number of society guidelines; however, screening rates remain suboptimal. Treatment includes a multidisciplinary approach focused on increasing physical activity, dietary interventions, and pharmacotherapy. This scientific statement outlines triglyceride metabolism and the pathophysiology of hypertriglyceridemia and provides an evidence-based review of the current literature on the screening, diagnosis and management of pediatric hypertriglyceridemia. This review highlights the need for additional research for effective lifestyle and pharmacotherapy for the treatment of pediatric hypertriglyceridemia.

Key Words: AHA Scientific Statements ■ adolescent ■ atherosclerosis ■ dietary therapy ■ exercise therapy ■ hypercholesterolemia ■ hypertriglyceridemia ■ pancreatitis

Pediatric hypertriglyceridemia, defined as elevated fasting plasma triglycerides levels for age,¹ commonly occurs with elevated non-high-density lipoprotein (HDL) cholesterol (HDL-C), insulin resistance (IR), and metabolic syndrome. It is increasingly recognized as obesity and metabolic syndrome have increased in prevalence.^{2–4} Hypertriglyceridemia affects an estimated 10% to 20% of US youths, with prevalence up to 24% to 60% among youths with elevated body mass index.^{5–9} Hypertriglyceridemia can lead to life-threatening acute pancreatitis in youths for the most severe cases; in others, hypertriglyceridemia increases the risk for atherosclerotic cardiovascular diseases (ASCVDs) such as myocardial infarction and stroke in adulthood.^{10–12} It is important to recognize and treat pediatric hypertriglyceridemia to mitigate pancreatitis and later ASCVD risk (Figure 1).¹²

CLASSIFICATION OF HYPERTRIGLYCERIDEMIA

Hypertriglyceridemia classification has evolved to address both ASCVD and pancreatitis risk, considering specific triglyceride levels and their management implications. In light of existing heterogeneous classification systems,^{13–18} guidance is needed to comprehensively address the dual risks of adult-onset ASCVD and acute pancreatitis in pediatric patients. It is important to note that management approaches differ for hypertriglyceridemia based on the triglyceride level.¹⁵ From a practical standpoint, the 2019 American Heart Association guidelines for high-risk pediatric patients classify hypertriglyceridemia as moderate (<400 mg/dL, <4.5 mmol/L), significant (400–999 mg/dL, 4.5–11.2 mmol/L), and severe (≥1000 mg/dL, ≥11.3

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Pediatric Hypertriglyceridemia

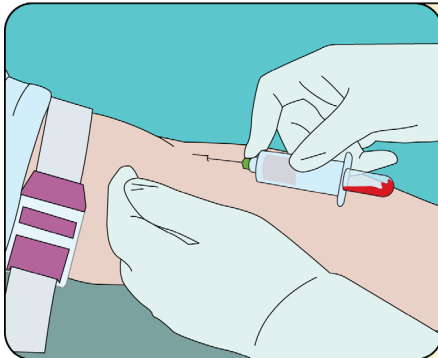


Causes

- Genetics

Secondary causes

- Obesity
- Endocrine disorders
- Drugs
- Other



Screening/diagnosis

- Universal lipid screening
- Fasting lipid panel

Treatment

- Dietary intervention
- Increase physical activity
- Pharmacotherapy

Complications: Increased risk of ASCVD Acute pancreatitis

Figure 1. Central illustration. ASCVD indicates atherosclerotic cardiovascular disease.

mmol/L) risk.¹⁹ At lower elevations, hypertriglyceridemia is driven primarily by endogenous synthesis (very-low-density lipoproteins [VLDL]), but when levels exceed 885 mg/dL (10 mmol/L), lipoprotein lipase (LPL) is less able to metabolize chylomicrons. Chylomicrons start to accumulate, significantly increasing the risk of pancreatitis. Consequently, for triglyceride levels >885 mg/dL (>10 mmol/L), interventions focus on reducing pancreatitis risk. In contrast, for triglyceride levels <885 mg/dL (<10 mmol/L), the primary concern is ASCVD risk due to elevated VLDL, which contains apolipoprotein (apo) B₁₀₀ and substantial cholesterol content.

In this scientific statement, we review the pathophysiology of hypertriglyceridemia, associated risks, and implications for management. We classify hypertriglyceridemia into discrete categories to address both pancreatitis and ASCVD risks considering pathophysiology and focus on practical management strategies for specific triglyceride thresholds. We have altered the severity classification and changed the cut points for the moderately and severely elevated levels compared with the 2019 “Cardiovascular Risk Reduction in High-Risk Pediatric Patients” scientific statement.¹⁹ These categories are shown in Table 1. Triglyceride levels <400 mg/dL (<4.5 mmol/L) highlight the importance of calculating non-HDL-C to assess ASCVD risk and to potentially consider

statin therapy in children ≥8 years of age. Moderately elevated triglyceride levels reflect thresholds for which fibrate therapy might be considered; severely elevated triglyceride levels (>885 mg/dL, >10 mmol/L) indicate levels for which chylomicrons begin to predominate and pancreatitis risk increases; and very severely elevated levels (>2000 mg/dL, >22.6 mmol/L) reflect the inflection point for significant increases in relative and absolute risk of acute pancreatitis.²⁰

TRIGLYCERIDE METABOLISM

Circulating triglyceride-rich lipoproteins (TRLs) include chylomicrons, VLDL, and remnant particles (eg, intermediate-density lipoprotein [IDL] and chylomicron remnants).¹⁶ Figure 2 provides an illustration of triglyceride metabolism. Triglycerides enter the circulation by ingestion (exogenous pathway) and endogenous production. Triglycerides within chylomicrons and VLDL are hydrolyzed by endothelium-bound LPL, which releases free fatty acids (FFAs) and leaves remnant particles. LPL function is enhanced by apo CII, apo AV, intracellular chaperone lipase maturation factor 1, and GPIHBP1 (glycosylphosphatidylinositol anchored HDL-binding protein 1). LPL function is downregulated by apo CIII and ANGPTL3 and ANGPTL4 (angiopoietin-like proteins 3 and 4).²¹

Table 1. Ideal, Borderline, and Elevated Triglyceride Levels in Youths: Diagnosis and Management

Classification	Acceptable	Mildly elevated	Moderately elevated	Severely elevated	Very severely elevated
Plasma TG concentration	≤130 mg/dL ≤1.47 mmol/L	131–400 mg/dL 1.48–4.5 mmol/L	401–885 mg/dL 4.53–10 mmol/L	886–2000 mg/dL 10.01–22.6 mmol/L	>2000 mg/dL ≥22.6 mmol/L
Predominant circulating lipoprotein	VLDL	VLDL	VLDL	VLDL and chylomicrons	Mainly chylomicrons
Key elements of lifestyle management	Promote routine health behaviors Reduce intake of sugar and refined carbohydrates, especially sugar-sweetened beverages	Engage in moderate to vigorous physical activity Manage overweight or obesity Reduce sugar to <5% of calories and reduce refined carbohydrates, especially sugar-sweetened beverages Reduce dietary fat to 25%–30% of daily kcal Discuss ASCVD risk	Engage in moderate to vigorous physical activity Manage overweight or obesity Reduce sugar to <5% of kcal and reduce refined carbohydrates, especially sugar-sweetened beverages Reduce dietary fat to 20%–25% of total kcal Discuss ASCVD and pancreatitis risk	Engage in moderate to vigorous physical activity Manage overweight or obesity Restrict dietary fat to 10%–15% of total daily kcal; this may equate to 15–20 g fat intake per day. Discuss pancreatitis risk	Engage in moderate to vigorous physical activity Manage overweight or obesity A “no-fat” or extremely low-fat diet is very effective in lowering TGs. Discuss pancreatitis risk
Key considerations in pharmacotherapy management	No pharmacotherapy needed	Pharmacotherapy is generally not needed. Calculate non-HDL-C; if persistently ≥145 mg/dL (≥3.8 mmol/L) despite lifestyle changes, consider statins in youth ≥8 y of age.	Consider treatment with fenofibrate or omega-3 FAs after trial of lifestyle management Calculate non-HDL-C; if persistently ≥145 mg/dL (≥3.8 mmol/L) despite lifestyle changes, consider statins in youth ≥8 y of age.	Consider treatment with fenofibrate or omega-3 FAs after trial of lifestyle management	These patients do not necessarily benefit from TG-lowering medication.
Primary goal of therapy	Prevent ASCVD	Prevent ASCVD	Prevent ASCVD	Prevent pancreatitis	Prevent pancreatitis

To convert triglycerides from mmol/L to mg/dL, multiply by 88.57. To convert non-HDL-C from mmol/L to mg/dL, multiply by 38.67.

ASCVD indicates atherosclerotic cardiovascular disease; FA, fatty acid; HDL-C, high-density lipoprotein cholesterol; TG, triglyceride; and VLDL, very-low-density lipoproteins.

The Exogenous Pathway

Digestion of dietary fat by intestinal lipases releases FFAs and monoglycerides, which are emulsified by bile salts and absorbed within micelles into enterocytes. There, FFAs are reincorporated into triglycerides through diacylglycerol O-acyltransferase and are fused by microsomal triglyceride transfer protein with cholesteryl ester (CE) and apo B₄₈, AI, CII, CIII, and E to form nascent chylomicrons.²¹ These traverse the lymphatics and enter the circulation through the thoracic duct. In peripheral tissues, endothelium-bound LPL hydrolyzes triglycerides within circulating chylomicrons to liberate FFAs for energy production and intracellular storage, and remnant particles undergo receptor-mediated hepatic endocytosis, involving primarily apo E-dependent mechanisms.

The Endogenous Pathway

Hepatocytes package cholesterol, CE, and triglycerides into VLDL. Hepatic diacylglycerol O-acyltransferase catalyzes the key step in triglyceride synthesis, and microsomal triglyceride transfer protein fuses triglycerides and CE with apo B₁₀₀, assembling VLDL in the endoplasmic reticulum. Once the VLDL particles are fully assembled and mature, they are secreted from the liver into the circulation. VLDL undergoes lipolysis by LPL, releasing

FFAs for tissue uptake while leaving behind IDL remnants. VLDL and IDL contain apo B₁₀₀, E, CII, and CIII. Circulating IDL undergoes further lipolysis that is partly mediated by hepatic lipase to form low-density lipoprotein (LDL) particles containing primarily apo B₁₀₀. Circulating remnant particles are cleared from plasma through receptor-mediated endocytosis facilitated by apo E, involving interactions with LDL receptor family members on the hepatocyte surface.

Chylomicron remnants, IDL, LDL, and VLDL are considered proatherogenic, whereas chylomicrons are thought to be too large to penetrate the arterial wall and therefore are not atherogenic.²² Hyperchylomicronemia is associated with annual risk of acute pancreatitis of 5% when triglycerides are >1000 mg/dL (>11.3 mmol/L) and up to 20% when triglycerides are >2000 mg/dL (>22.6 mmol/L).^{22,23} Emerging evidence also emphasizes the importance of the temporal dynamics of hyperchylomicronemia in modulating pancreatitis risk.²⁴ Chylomicron particles are lipotoxic within the pancreatic microvasculature; their postprandial accumulation can lead to ischemia and localized inflammation attributable to release of toxic intermediate lipid and phospholipid species. Thus, not only fasting triglyceride levels but also the duration and postprandial persistence of chylomicron-rich lipoproteins modulate the risk of acute pancreatitis.

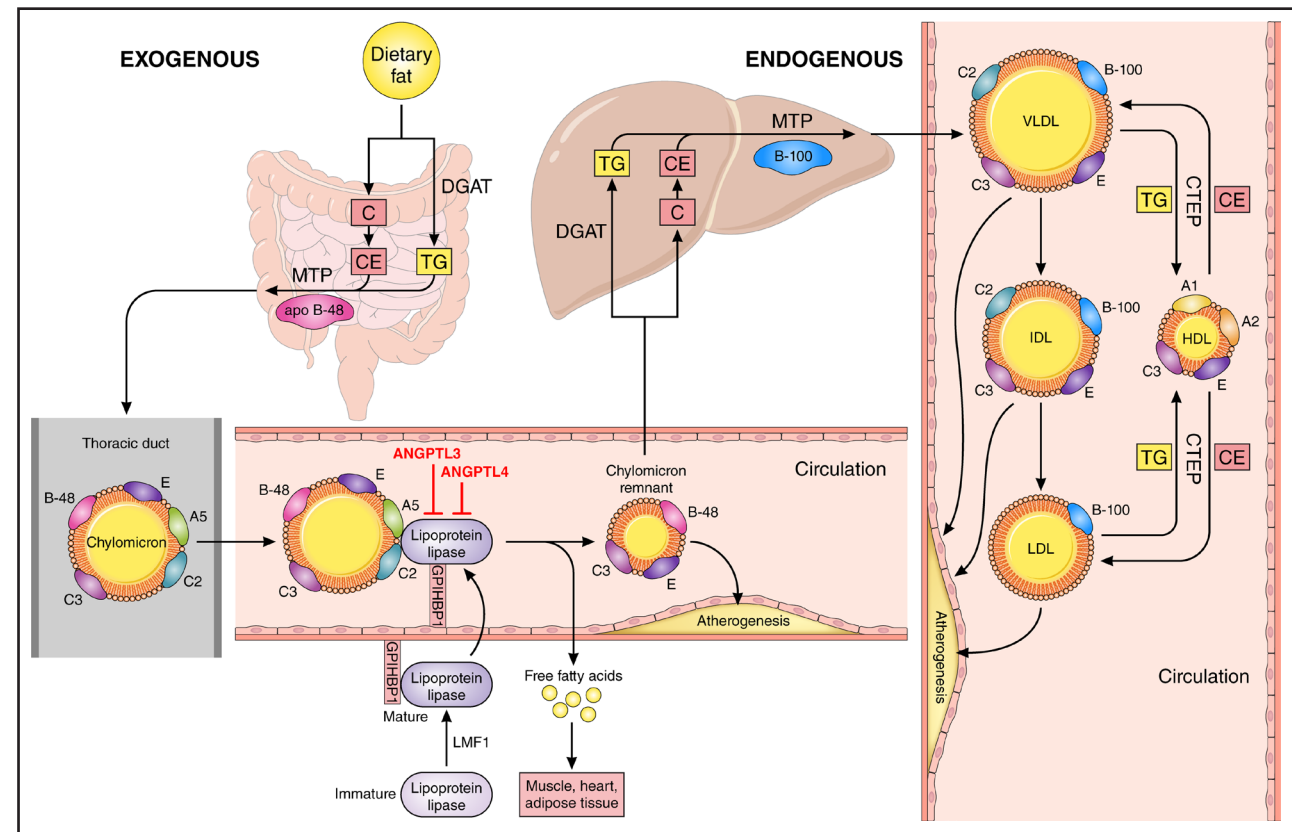


Figure 2. Triglyceride metabolism.

The exogenous pathway involves the intestinal absorption of dietary fats and the subsequent formation of triglycerides (TGs) from free fatty acids (FFAs) through the enzyme diacylglycerol O-acyltransferase (DGAT). These TGs, with the help of microsomal triglyceride transfer protein (MTP), are combined with cholesterol esters (CEs) and apolipoprotein (apo) B₄₈, A1, CII, CIII, and E to form nascent chylomicrons. After passage through lymphatics, chylomicrons in plasma undergo hydrolysis by lipoprotein lipase (LPL), releasing FFAs for energy use in peripheral tissues. Chylomicron remnants are taken up by the liver through low-density lipoprotein (LDL) receptor-related protein. The chaperone protein lipase maturation factor 1 (LMF1) ensures correct folding and posttranslational processing of LPL. After LPL is secreted into the interstitium by adipocytes and myocytes, glycosylphosphatidylinositol-anchored high-density lipoprotein-binding protein 1 (GPIHBP1) transports and anchors it to the luminal surface of capillaries, facilitating TG hydrolysis. Key regulators of LPL-induced hydrolysis include activators (apo CII and apo AV), inhibitors (apo CIII, ANGPTL3 [angiopoietin-like protein 3], and ANGPTL4 [angiopoietin-like protein 4]), and essential cofactors (LMF1 and GPIHBP1). In the endogenous pathway, TGs formed from FFA through DGAT, along with cholesterol and CE, are assembled by MTP with apo B₁₀₀ into very-low-density lipoprotein (VLDL) in hepatocytes. Circulating VLDL particles are hydrolyzed by LPL, releasing FFAs and forming VLDL remnants, also known as intermediate-density lipoproteins (IDL), which are subsequently hydrolyzed by hepatic lipase to form low-density lipoprotein (LDL). Cholesteryl ester transfer protein (CETP) facilitates the transfer of CE from high-density lipoprotein (HDL) to LDL and VLDL and of TGs from LDL and VLDL to HDL. In contrast to chylomicrons, which are not considered to be proatherogenic, VLDL, chylomicron remnants, and IDL are proatherogenic.

Pathophysiology of IR, LPL Dysfunction, Hypertriglyceridemia, and a Proatherogenic State

Hypertriglyceridemia results from increased production or impaired clearance of TRLs or both. In cases of mild hypertriglyceridemia, the key contributors are a diet high in refined carbohydrates and sugars, excess calories, sedentary lifestyle, and IR. Excessive dietary carbohydrates, in the background of IR, lead to increased de novo lipogenesis. IR enhances adipocyte lipolysis, increasing the release of FFAs into the plasma. This increased FFA flux fuels hepatic overproduction of VLDL. In addition, IR impairs the activity of LPL, leading to reduced clearance of TRLs.^{25–27} The accumulation of TRLs promotes

remodeling of lipoproteins through the action of CE transfer protein, which facilitates the exchange of triglycerides from TRLs into LDL particles in exchange for CEs. This lipid exchange produces triglyceride-enriched LDL particles, which are subsequently hydrolyzed by hepatic lipase to form proatherogenic small dense LDL particles,²⁸ which show increased susceptibility to oxidation and greater arterial wall penetration.

When hypertriglyceridemia progresses to moderate levels, it is often associated not only with excessive endogenous synthesis of VLDL but also with impaired lipolysis of TRLs. This dual dysfunction leads to the accumulation of both VLDL and chylomicrons in circulation. Both large VLDL and chylomicrons compete for clearance through LPL. Severe hypertriglyceridemia often

results from more profound dysfunction of triglyceride clearance mechanisms. These can include impaired LPL activity or the genetic absence of LPL, leading to hyperchylomicronemia.

All apo B₁₀₀-containing lipoprotein particles, including VLDL, IDL, LDL, and lipoprotein(a), are atherogenic. The calculation of non-HDL-C (total cholesterol minus HDL-C) from a lipid panel can be used as a surrogate marker to estimate the burden of these particles. However, it is important to distinguish remnant cholesterol (typically defined as the cholesterol content of VLDL and IDL) from non-HDL-C. Remnant cholesterol specifically reflects the cholesterol carried by TRLs, which are pro-inflammatory and penetrate the arterial wall more readily than larger particles. In contrast, non-HDL-C encompasses both remnant cholesterol and cholesterol carried by LDL and lipoprotein(a). Both non-HDL-C and remnant cholesterol appear to be more predictive than LDL cholesterol (LDL-C) alone for ASCVD risk in individuals with hypertriglyceridemia.¹² Non-HDL-C is readily calculated from a standard lipid panel, and reference ranges for non-HDL-C and apoB in youths are available¹; measurement of apoB can be considered. Guidance on lipoprotein(a) testing in youths is also available.²⁹

GENETICS OF HYPERTRIGLYCERIDEMIA

Hypertriglyceridemia has complex genetic architecture.³⁰ A summary of genetic causes of hypertriglyceridemia in youths is given in Table 2. Mild to moderate hypertriglyceridemia involves primarily elevated VLDL²¹ and affects 1 in 10 to 20 adults, with an unclear but likely increasing prevalence in children.³¹ Patients have increased polygenic risk, heterozygous rare variants, or both.³² Clinically, these patients have mildly to moderately elevated fasting triglycerides (eg, 200–500 mg/dL, 2.26–5 mmol/L), low HDL-C, and normal or borderline LDL-C. They may have a family history of ASCVD, fatty liver, or IR. Although they may not experience overt symptoms, they carry increased long-term cardiometabolic risk. With management of lifestyle factors (ie, diet, physical activity, and alcohol intake), patients can often control triglyceride levels, although periods of severe hypertriglyceridemia may occur,³³ especially in response to metabolic stressors such as uncontrolled diabetes, acute illness, or medications (eg, corticosteroids, estrogen therapy).

Multifactorial chylomicronemia syndrome (MCS) affects 1 in 400 to 500 adults, but its prevalence in youths is unclear. MCS can arise from heterozygous pathogenic variants in lipolysis genes in 15% to 20% of cases, polygenic risk in >50% of cases, caused by the cumulative effect of multiple triglyceride-raising single nucleotide polymorphisms, or both.²⁷ Polygenic risk is quantified by a polygenic score.³⁴ MCS follows a probabilistic genotype-phenotype relationship; genetic factors increase the likelihood but do not guarantee

development of hypertriglyceridemia, which is also influenced by secondary factors such as diet, obesity, diabetes, alcohol use, hypothyroidism, and certain medications. Autoantibodies against GPIHBP1 can cause acquired intermittent hyperchylomicronemia.³⁵

Combined hyperlipidemia affects 1 in 50 to 100 adults and is linked to early ASCVD. Combined hyperlipidemia is characterized by elevations in total cholesterol and triglycerides, driven largely by elevated VLDL cholesterol along with elevated LDL-C. Consequently, apoB and non-HDL-C levels are often elevated, reflecting the increased burden of atherogenic particles.²¹ Combined hyperlipidemia has a polygenic pathogenesis similar to MCS³⁶ and is associated with obesity, IR, and type 2 diabetes.²¹

Dysbetalipoproteinemia affects 1 in 1000 to 3000 adults,²¹ although childhood cases have been reported.³⁷ Patients have elevated total cholesterol and triglycerides attributable to IDL and chylomicron remnant accumulation.²¹ Most patients are homozygous for the apo E2 isoform, but nongenetic factors such as obesity and diabetes are usually present. Rarely, dominant-negative *APOE* variants that encode a variant protein that interferes with the normal apo E may cause this condition.²¹

Familial chylomicronemia syndrome (FCS) is the archetype of genetically severe or very severe hypertriglyceridemia, usually with onset in youths.^{38–40} FCS has a prevalence of 1 in 100 000 to 1 000 000 and results from biallelic pathogenic variants in *LPL* (60%–80% of cases), *GPIHBP1*, *APOA5*, *APOC2*, or *LMF1*.^{38,39,41} These pathogenic variants abolish lipolytic activity, resulting in chylomicron accumulation and hypertriglyceridemia.^{21,38,39,41}

Transient childhood hypertriglyceridemia is caused by biallelic pathogenic variants in *GPD1*, leading to glycerol-3-phosphate dehydrogenase 1 deficiency.⁴² Hypertriglyceridemia is due to increased VLDL and is associated with fatty liver but is typically transient, with triglyceride levels normalizing by 10 years of age, possibly as a result of compensatory enzymatic pathways.³⁹

Hypertriglyceridemia is also seen in ≈30% of patients with lysosomal acid lipase deficiency, a disorder of intrahepatic cholesterol metabolism caused by biallelic pathogenic variants in the *LIPA* gene. Lysosomal acid lipase deficiency encompasses a spectrum of disease severity, with Wolman disease presenting as a severe form in infancy and CE storage disease as a less severe form that arises later in life.

Heritable lipodystrophies can lead to severe IR and hypertriglyceridemia.⁴³ Two primary forms are familial partial lipodystrophy, a dominant condition caused by pathogenic variants in *LMNA* or *PPARG*, and congenital generalized lipodystrophy, a recessive disorder caused by pathogenic variants in *AGPAT2* or *BSCL2*. Patients with lipodystrophy who develop severe or very severe hypertriglyceridemia may benefit from metreleptin and emerging agents such as inhibitors of *APOC3*.⁴³

Table 2. Genetic Causes of Pediatric Hypertriglyceridemia

Key lipid cut point*	Name of condition	Synonym(s)	Clinical features	Inheritance pattern	Prevalence	Gene name/ chromosome	Comments
TG >10 mmol/L or ≥1000 mg/dL	FCS	Type 1 FLL HLP, LPL deficiency, apo CII deficiency, monogenic chylomicronemia	Failure to thrive, lipemia retinalis, hepatosplenomegaly, eruptive xanthomas, abdominal pain, nausea, vomiting, pancreatitis	Mono-genic autosomal recessive	1:100 000–1 000 000	<i>LPL</i> /8p22	≈80% of FCS cases are LPL deficiency with biallelic variants; most patients are compound heterozygotes (ie, 2 different variants) with clinical severity equal to that of simple homozygotes (ie, 2 copies of the identical variant)
						<i>GPIHBP1</i> /8q24.3	3%–6% of FCS cases
						<i>APOA5</i> /11q23	2%–5% of FCS cases
						<i>APOC2</i> /19q13.2	Classical apo CII deficiency, 2%–5% of FCS cases
						<i>LMF1</i> /16p13.3	2%–5% of FCS cases
TG >5 mmol/L or ≥500 mg/dL	Transient infantile HTG		Failure to thrive, abdominal pain	Mono-genic autosomal recessive	<1:500 000–1 000 000	<i>GPD1</i> /12q12	VLDLs are mainly elevated; lipid profiles improve as the child grows.
TG >10 mmol/L or ≥1000 mg/dL	Congenital generalized lipodystrophy	Berardinelli-Seip syndrome	Complete absence of subcutaneous fat, hepatosplenomegaly, intramyocellular fat, pancreatitis	Mono-genic autosomal recessive	<1:500 000–1 000 000	<i>AGPAT2</i> /9q34.3	These are the 2 most common genetic causes of congenital generalized lipodystrophy; other very minor genetic loci have been reported in as few as 1 family.
						<i>BSCL2</i> /1q12.3.3	
TG >5 mmol/L or ≥500 mg/dL	Partial lipodystrophy	Dunnigan-type familial partial lipodystrophy	Loss of subcutaneous fat on limbs and gluteal region, with excess on head and neck, trunk, abdomen, and viscera	Mono-genic autosomal dominant	1:60 000–300 000	<i>LMNA</i> /1q22	These are the 2 most common genetic causes of inherited partial lipodystrophy; other very minor genetic loci have been reported in as few as 1 family.
						<i>PPARG</i> /3p25.2	
TG >10 mmol/L or ≥1000 mg/dL	MCS	Type 5 FLL HLP, familial hypertriglyceridemia (severe form), mixed hyperlipoproteinemia	A, B, or both A, Biochemical disturbance alone B, lipemia retinalis, hepatosplenomegaly, eruptive xanthomas, abdominal pain, nausea, vomiting, pancreatitis, increased ASCVD risk	Complex	1:400–600	Not applicable (polygenic)	1, 2, or both: (1) Heterozygosity for a variably expressed pathogenic variant in 1 of FCS genes— <i>LPL</i> , <i>GPIHBP1</i> , <i>APOA5</i> , <i>APOC2</i> , and <i>LMF1</i> —in 15%–20% of affected subjects; (2) Polygenic predisposition associated with a high SNP score in 35–50% of subjects
TG 2.0–9.9 mmol/L or TG 200–1000 mg/dL	Mild to moderate HTG	Type 4 FLL HLP, familial hypertriglyceridemia (mild form)	Typically no clinical features	Complex	1:20–40	Not applicable (polygenic)	Similar genetic background as MCS; secondary factors play an important role
TC >6 mmol/L and TG >5 mmol/L or TC ≥250 mg/dL and TG ≥500 mg/dL	Dysbetalipoproteinemia	Type 3 FLL HLP, remnant hyperlipidemia, familial dysbetalipoproteinemia, board beta disease	Usually none, palmar xanthomas, tuberos xanthomas	<i>APOE</i> E2/E2 or dominant rare variant of <i>APOE</i>	1:1000–3000	<i>APOE</i> /19q13 but also with an underlying polygenic component	Most typically presents in adulthood Heterozygous <i>APOE</i> rare pathogenic variants or homozygosity for common E2 isoform are predisposing factors, with similar genetic background as MCS.
TC >6 mmol/L and TG >3 mmol/L or TC >250 mg/dL and TG >250	Combined hyperlipidemia	Type 2b FLL HLP, familial combined hyperlipidemia	Biochemical disturbance alone	Polygenic	1:50–100	Not applicable (polygenic)	No clear inheritance pattern. Apo B level is also elevated, and HDL-C is depressed. Similar genetic background to MCS; secondary factors play an important role

Apo indicates apolipoprotein; ASCVD, atherosclerotic cardiovascular disease; FCS, familial chylomicronemia syndrome; FLL, Frederickson-Levy-Lees; HDL-C, high-density lipoprotein cholesterol; HLP, hyperlipoproteinemia; HTG, hypertriglyceridemia; LPL, lipoprotein lipase; MCS, multifactorial chylomicronemia syndrome; SNP, single-nucleotide polymorphism; TC, total cholesterol; TG, triglyceride; and VLDL, very-low-density lipoprotein.

*Lipid cut points are rounded to reflect clinically relevant values for the particular units based on convention and published literature. Values as shown for the respective unit systems can be used interchangeably for practical purposes.

SECONDARY CAUSES OF HYPERTRIGLYCERIDEMIA

Hypertriglyceridemia can result from various secondary causes, including those related to IR and LPL dysfunction, namely dietary factors, sedentary lifestyle, diabetes, and metabolic syndrome, with IR related to obesity or polycystic ovarian syndrome. Other secondary causes include medications, pregnancy, kidney disease, and alcohol use. Most of these conditions are usually associated with mild to moderate hypertriglyceridemia related to increased triglyceride synthesis (VLDL overproduction) and impaired triglyceride hydrolysis or an MCS-like picture as described previously. A more comprehensive list of conditions that can cause secondary hypertriglyceridemia is given in Table 3.

Obesity with associated IR is the leading cause of secondary hypertriglyceridemia in youths,⁴⁴ and its severity correlates with the degree of hypertriglyceridemia.⁴⁵ Diets high in processed, calorie-dense foods with a high carbohydrate content,⁴⁶ high glycemic index, or high fat content and consumption of sugar-sweetened beverages rich in sucrose or fructose can lead to excess calorie intake and IR, which increase endogenous hepatic VLDL production and impair lipolysis of TRLs. Limited physical activity, particularly when associated with obesity, contributes to dyslipidemia.

Pregnancy can be associated with hypertriglyceridemia, particularly in the third trimester as hepatic synthesis of VLDL increases physiologically.⁴⁷ A key hormonal change is the increase in human placental lactogen, which is critical for fetal growth and maternal lactation adaptations, but it is mediated by IR with consequent dyslipidemia.

Acute and chronic kidney conditions such as nephrotic syndrome and kidney failure often result in hypertriglyceridemia with heterogeneous duration, severity, and clinical impact. Nephrotic syndrome causes hypertriglyceridemia through various mechanisms, including reduced hepatic lipase expression and activity, impaired VLDL and chylomicron clearance, reduced LPL expression and activity, and defective endothelial binding related to altered apo E content and increased hepatic secretion of key proteins.^{48–50} Although nephrotic syndrome–related hypertriglyceridemia may not require pharmacological treatment because triglyceride levels can return to normal when a patient is in remission, it may increase ASCVD risk, especially with concurrent hypercoagulability commonly seen in these patients.⁴⁸ Glomerulonephritis, from a range of primary and secondary pathogenesises, causes hypertriglyceridemia in part as a result of the impact of the proinflammatory state on lipolysis, although the precise kidney-specific mechanisms are incompletely understood. Youths with chronic kidney disease have an increased risk of hypertriglyceridemia regardless of their underlying renal pathology, which further increases their

ASCVD risk.^{51,52} Kidney failure is especially complicated because triglyceride-lowering medications are often dependent on renal clearance and are not dialyzable, possibly leading to excessive toxicity such as rhabdomyolysis.⁵³ Hypertriglyceridemia has also been implicated in contributing to incident kidney disease and kidney disease progression.^{54,55}

Metabolic dysfunction–associated steatotic liver disease is linked to increased de novo lipogenesis driven by excessive dietary carbohydrate intake, enhanced secretion, and impaired clearance of VLDL. Chronic inflammatory conditions can raise triglyceride levels through cytokine-mediated suppression of lipolysis.

SCREENING AND DIAGNOSIS OF HYPERTRIGLYCERIDEMIA

Multiple guidelines endorsing screening and diagnosis of lipid disorders in youths have been published over the past few decades, the recent United States Preventive Services Task Force recommendations^{56,57} notwithstanding ([Supplemental Material Table](#)). Most guidelines have focused on screening for high cholesterol, with hypertriglyceridemia being a secondary variable of interest. Universal cholesterol screening for all youths 9 to 11 and 17 to 21 years of age is aimed primarily at diagnosing familial hypercholesterolemia; is recommended by the National Heart, Lung, and Blood Institute; and is endorsed by the American Academy of Pediatrics,¹ American Heart Association,^{17,19} American College of Cardiology,¹⁷ and National Lipid Association.⁵⁸ Screening of youths beyond those with overweight or obesity is supported in part by population-representative data from 1988 to 2016 showing that 26% to 55% of youths with high non-HDL-C or low HDL-C have neither overweight nor obesity.⁴ Therefore, testing only youths with overweight or obesity would miss these cases. Selective screening has also been recommended for youths with specific ASCVD risk factors such as obesity,⁵⁹ kidney disease,^{60,61} diabetes,^{62,63} and hypertension.⁶⁴ This combined approach may also detect hypertriglyceridemia, a condition often linked to metabolic or lifestyle-related risks.

Cut points for acceptable and elevated triglycerides are shown in Table 1. Although a nonfasting lipid panel is more convenient and is often used for universal screening, if non-HDL-C is ≥ 145 mg/dL (≥ 3.8 mmol/L) or the HDL-C level is < 40 mg/dL (< 1 mmol/L), then physicians can consider a repeat fasting lipid panel. Because triglyceride levels are sensitive to prandial state, 8 to 12 hours of fasting before confirmatory testing can be considered.¹ Treatment decisions are generally based on fasting triglyceride levels. However, studies show that in adults elevated postprandial hypertriglyceridemia is associated with IR and metabolic syndrome, and postprandial triglyceride measurement may better predict adult

Table 3. Causes of Secondary Hypertriglyceridemia in Youths

Obesity and obesity-related medical conditions
Body mass index ≥85th percentile
Metabolic syndrome
Endocrine-related disorders
Inadequately controlled type 1 or type 2 diabetes
Hypothyroidism
Hypercortisolemia
Lipodystrophy (primary genetic or acquired)
Drug exposure
Alcohol use
Retinoids, including isotretinoin
Anabolic steroids
Glucocorticoids
Oral estrogens
Estrogen-receptor blockers
Antipsychotic medications, including first- and second-generation antipsychotics
Tricyclic antidepressants
Selective serotonin reuptake inhibitors
Bile acid sequestrants and bile acid-binding resins
β-Blockers
Diuretics, including thiazide and loop diuretics
Protease inhibitors
Immunosuppressants, including rapamycin, sirolimus, tacrolimus, cyclosporine, azathioprine, mycophenolate mofetil
Pegylated conjugate of L-asparaginase
Tamoxifen
Cyclophosphamide
Rosiglitazone
Type 1 kinase inhibitors
Other
Pregnancy (especially third trimester)
Acute and chronic infection
Glycogen storage disorders
Renal disease
Autoimmune/chronic inflammatory diseases
Parenteral nutrition/intravenous lipid emulsions
Other liver diseases

ASCVD events than fasting triglyceride levels, although objective standards for obtaining nonfasting triglyceride levels have not been established.^{65–67}

Once hypertriglyceridemia is diagnosed, evaluation for secondary causes and other ASCVD risk factors can be considered even if a genetic cause is suggested (see diagnosis and evaluation of hypertriglyceridemia for youths in Figure 3). Most secondary causes can be ruled out with a careful history and physical examination, including family history and measurement of height, weight, and blood pressure. Treatment and optimization

of secondary causes of hypertriglyceridemia can be considered while lifestyle management is initiated. Evaluation for other ASCVD risk factors such as hypertension, IR, diabetes, and metabolic dysfunction–associated steatotic liver disease can also be considered.

MANAGEMENT OF HYPERTRIGLYCERIDEMIA

Previous recommendations for managing hypertriglyceridemia in youths were based on limited pediatric-specific guidelines and extrapolation from adult guidelines and outcomes-based data.^{1,15,17,18} The cornerstones of hypertriglyceridemia management involve lifestyle modification consisting of targeted dietary changes, increased moderate to vigorous physical activity, and management of pediatric overweight or obesity. Pharmacotherapy recommendations have historically been based on the dominant circulating lipoprotein expected at each severity level (Table 1). Most mild to moderate hypertriglyceridemia can be managed in the primary care setting with dietary and lifestyle intervention. If pharmacotherapy is considered, referral to a pediatric lipidologist, pediatric preventive cardiologist, or pediatric endocrinologist can be considered, depending on local access to these services.

Lifestyle Modification for Mild to Moderate Hypertriglyceridemia

After any secondary causes of hypertriglyceridemia have been identified and treated or optimized, lifestyle management is the first-line therapy for all patients with persistent hypertriglyceridemia^{11,68} and, when adopted at a younger age, can help reduce lifetime ASCVD risk.⁶⁹ For mild to moderate hypertriglyceridemia, lifestyle changes can reduce triglycerides by 27%, and 36% of affected youths can normalize their triglyceride levels.^{16,70} Lifestyle management includes a family-centered approach and is best carried out with a registered dietitian with training and experience in pediatric dyslipidemia. Patients and caregivers can follow foundational dietary patterns outlined in the 2020 to 2025 Dietary Guidelines for Americans⁷¹ emphasizing eating more fruits and vegetables, lean proteins, plant-based proteins, low-fat dairy or soy products, whole grains, and oils rich in monounsaturated and polyunsaturated fats. Special attention can be paid to lowering consumption of refined carbohydrates and sugars.

For youths with mild hypertriglyceridemia, physicians can consider the Cardiovascular Health Integrated Lifestyle Diet 2-TG dietary approach. Interventions include a lower-fat diet (25%–30% of total kilocalories from fat, ≤7% from saturated fat, and ≈10% from monounsaturated

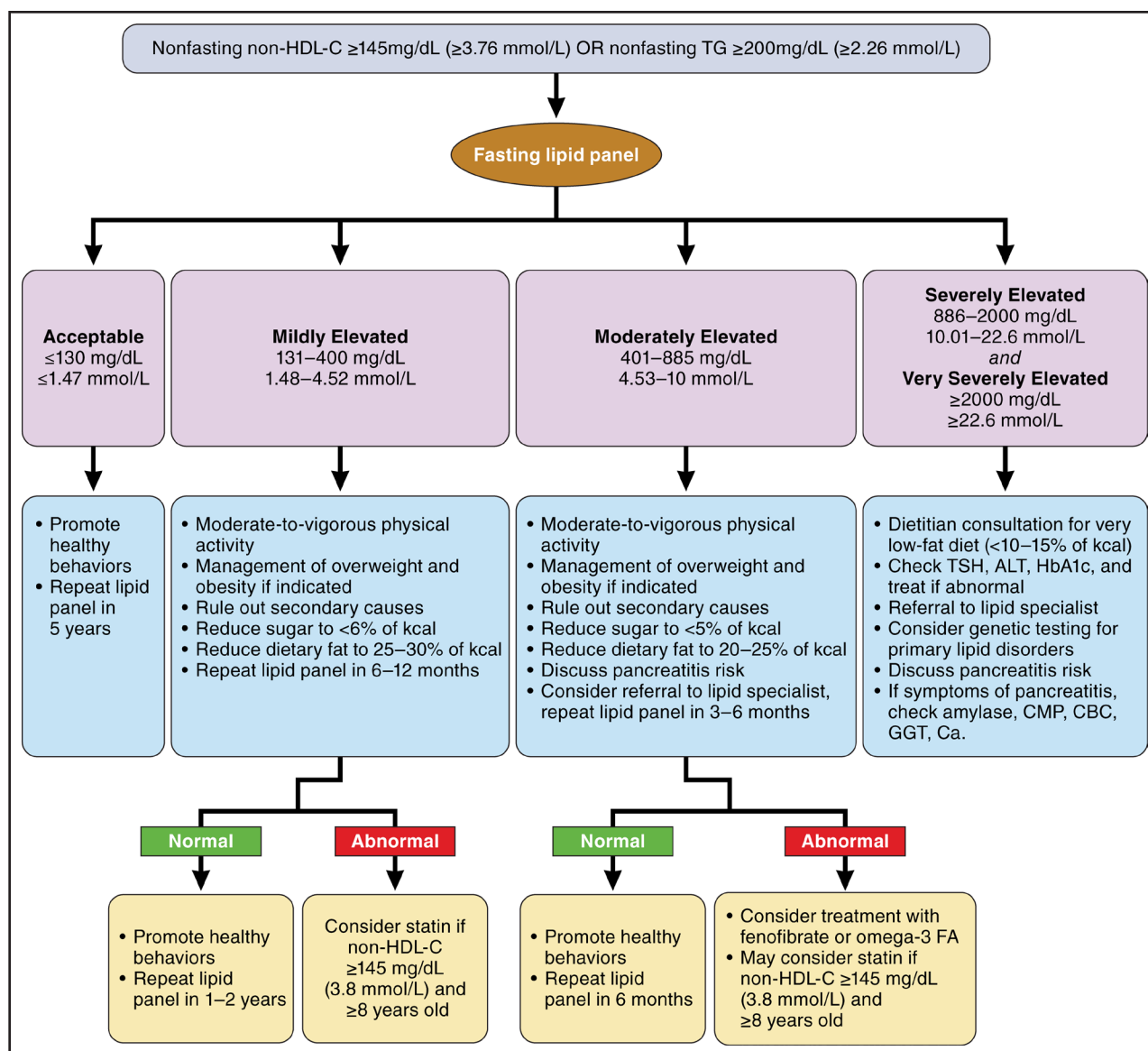


Figure 3. Management approach to pediatric hypertriglyceridemia.

ALT indicates alanine aminotransferase; CBC, complete blood count; CMP, complete metabolic panel; FA, fatty acid; GGT, gamma-glutamyl transferase; HbA1c, hemoglobin A1c; HDL-C, high-density lipoprotein cholesterol; TG, triglyceride; and TSH, thyroid-stimulating hormone.

fat, plus avoidance of *trans* fat), decreased sugar intake, and increased dietary fish rich in omega-3 fatty acid (FA; depending on age, 1–4 oz of fatty fish such as salmon, mackerel, cod, or tuna twice weekly).¹ Reducing total intake of refined carbohydrates, replacing simple carbohydrates with complex carbohydrates such as whole grains, and potentially decreasing kilocalorie intake are especially important given the major pathophysiological roles of obesity and IR in hypertriglyceridemia. This can be challenging because insulin and sugar activate dopaminergic pathways.⁷² The patient and family can consider avoidance strategies focused on increasing lean protein and cruciferous vegetable intake, which has the additional benefit of increasing dietary fiber intake, which further reduces triglyceride levels.^{73,74} As triglyceride

concentrations rise to the level of moderate hypertriglyceridemia, further dietary fat restriction to 20% to 25% of total daily kilocalorie intake from fat can be considered. The reason is that, at triglyceride levels exceeding 400 mg/dL (4.5 mmol/L), there is associated impairment in lipolysis of circulating fat.⁶⁸

Physical activity should be a regular part of a child's lifestyle and is a core tenet in multiple pediatric guidelines. Guidelines suggest that youths engage in 60 minutes of moderate to vigorous physical activity daily with muscle- and bone-strengthening activity 3 times a week.⁷¹ There are numerous technical definitions of moderate to vigorous physical activity, but experientially, the child should be sweating and mildly breathless and feel their heart beating quickly. They should be unable to sing a song

fluently or speak a complete sentence without catching their breath. Regular physical activity has been linked to lower body mass index, improved bone health, increased insulin sensitivity, and lower triglycerides.⁷⁵

Lifestyle Modification for Severe Hypertriglyceridemia

For severe and very severe hypertriglyceridemia, treatment initially targets reduction of acute pancreatitis risk. Dietary intervention is the mainstay of therapy and focuses on fat reduction to 10% to 15% of total daily calories (equivalent to ≈ 15 –20 g fat per day or less) when triglycerides are in the severely elevated range and a “very low or no-fat” diet at the very severely elevated range. This highly restrictive diet requires careful attention to ensure adequate intake of fat-soluble vitamins and essential fatty acids, especially for people with FCS who require lifelong fat restriction. Medium-chain triglyceride oil may be incorporated to provide additional calories because medium-chain triglycerides are absorbed directly into the portal circulation without requiring chylomicron formation or LPL activity, thus avoiding significant postprandial triglyceride elevation.⁷⁶ Recommendations for medium-chain triglyceride oil use should be individualized, keeping in mind that each tablespoon of medium chain triglyceride oil contains 14 g saturated fat and ≈ 126 calories, underscoring the need for careful dietary planning with consultation with registered dietitians with experience in lipid disorders. People with MCS should eliminate added simple carbohydrates and avoid *trans* and saturated fats. Abstinence from alcohol is essential.

Pharmacotherapy for Hypertriglyceridemia

There are no US Food and Drug Administration–approved treatments for hypertriglyceridemia in youths. Consultation with a pediatric lipid specialist should be considered when considering and before initiating pharmacotherapy. The primary indications for pharmacotherapy are (1) acquired or secondary hypertriglyceridemia with triglycerides persistently ≥ 400 mg/dL (≥ 4.5 mmol/L) that is refractory to lifestyle management, treatment of an underlying causal condition, or both and (2) primary genetic hypertriglyceridemia with high risk for pancreatitis.

For youths in the first group, the risk of pancreatitis is low; the main goal of therapy is to lower ASCVD risk, primarily through lowering apoB and non-HDL-C levels. First-line treatment options include omega-3 FA, fenofibrate, and, if non-HDL-C is persistently ≥ 145 mg/dL (3.8 mmol/L), a statin (for youths ≥ 8 years of age). The role of omega-3 FA in reducing triglycerides and preventing ASCVD is unclear. Two clinical trials of omega-3 FA therapy in children have not shown a significant reduction in triglyceride levels.^{77,78} In a large primary prevention trial in adults, combined docosahexaenoic acid/

eicosapentaenoic acid (EPA) supplementation did not reduce ASCVD events except in certain subpopulations.⁷⁹ An EPA supplementation trial in adults with prior ASCVD or diabetes did demonstrate ASCVD event protection, but this protection appeared to be independent of triglyceride change,⁸⁰ and questions have been raised about adverse cardiovascular effects from the placebo used in this trial.⁸¹ Another trial in adults using omega-3 carboxylic acid was terminated early because of futility.⁸² Omega-3 FAs are available in prescription formulations as combined docosahexaenoic acid/EPA and EPA-only formulations, as well as various over-the-counter supplements. Prescription formulations generally have high cost and limited coverage from insurance, whereas supplement formulations are underregulated and have wide variability in terms of docosahexaenoic acid and EPA content. Side effects from omega-3 FAs include a fish-like aftertaste or smell in sweat (if the product is derived from fish oil), burping, nausea, or bloating. A meta-analysis did not find any increased bleeding risk associated with combined docosahexaenoic acid/EPA, but very high-dose EPA was associated with a slight increase in bleeding risk.⁸³

For youths with no underlying genetic cause of their hypertriglyceridemia, fibrates can be used off-label in consultation with a pediatric lipid specialist. Generally, fenofibrate is preferred over gemfibrozil because of the lower risk of myositis and fewer drug-drug interactions with statins.¹⁷ Adverse effects from fibrates are rare in youths but include myalgias, abdominal pain, diarrhea, and constipation. Fibrates can raise transaminase levels, creatinine phosphokinase, and serum creatinine. Statins are the first-line treatment for adults ≥ 40 years of age with hypertriglyceridemia who are at elevated ASCVD risk¹⁷ but are not generally used to treat pediatric hypertriglyceridemia without concomitant elevated LDL-C. They can be considered for youths ≥ 8 years of age with hypertriglyceridemia who also have non-HDL-C ≥ 145 mg/dL (≥ 3.8 mmol/L). This consideration of statin therapy for pediatric patients is based on expert opinion. Statin monotherapy does not significantly reduce triglyceride levels; studies in adults show reductions of 10% to 20% from baseline.^{84–86} Niacin is not recommended for pediatric patients because of its limited efficacy in lowering triglycerides and limited safety data.

To treat hospitalized youths with very severe hypertriglyceridemia, avoiding enteral feeding with intravenous fluid replacement and, if hyperglycemia is present, administering insulin can reduce triglycerides by 80% within 24 hours.^{39,87} However, in normoglycemic individuals with severe hypertriglyceridemia, there is limited evidence to support the use of insulin, and its administration carries a risk of hypoglycemia. Insulin administration in youths without diabetes is generally done with concurrent dextrose and must be approached extremely carefully to avoid iatrogenic hypoglycemic seizures.⁸⁸ Plasmapheresis is rarely, if ever, necessary and

is reserved for the critically ill patient with very severe hypertriglyceridemia.⁸⁹

Newer agents developed to treat diabetes and obesity, namely glucagon-like peptide-1 receptor agonists and sodium-glucose cotransporter-2 inhibitors, may play a role in the treatment of pediatric hypertriglyceridemia related to diabetes, IR, or obesity but are not approved by the US Food and Drug Administration for hypertriglyceridemia treatment in youths. In adults with⁹⁰ and without⁹¹ type 2 diabetes, semaglutide (a glucagon-like peptide-1 agonist) reduced triglyceride levels and major adverse cardiac event risk compared with placebo. A pediatric semaglutide trial showed a 28.4% reduction in triglycerides in the treatment group compared with a 2.6% increase in the placebo group.⁹² In a similar trial, liraglutide (another glucagon-like peptide-1 agonist) did not reduce triglycerides.⁹³ Empagliflozin and dapagliflozin, both sodium-glucose cotransporter-2 inhibitors, are approved by the US Food and Drug Administration for the treatment of type 2 diabetes in youths, but their impact on triglyceride levels in youths is not clear. Studies in adults show that sodium-glucose cotransporter-2 inhibition lowers triglyceride levels but increases TC, HDL-C, and LDL-C levels.⁹⁴

For youths in the second group, those with primary genetic hypertriglyceridemia at risk for pancreatitis, clarifying the genetic triglyceride disorder will guide management. Generally, adherence to a low fat or “no-fat” diet is the mainstay of treatment. There is 1 encouraging trial of lomitapide in adults with FCS,⁹⁵ although its use should be limited to lipid specialists. Inhibitors of *APOC3* such as olezarsen⁹⁶ and plzasiran⁹⁷ show great promise in reducing triglyceride and pancreatitis events in FCS; both have been approved by the US Food and Drug Administration for adults with FCS.

POPULATION-BASED APPROACHES TO PREVENT HYPERTRIGLYCERIDEMIA

Approximately 1 in 5 youths has abnormal cholesterol, many of whom also have hypertriglyceridemia.⁴ Youths who self-report their race and ethnicity as Mexican American or non-Hispanic Black have lower mean cardiovascular health scores and social determinants of health scores compared with non-Hispanic White youths.⁹⁸ Hypertriglyceridemia is highly associated with excess body weight.⁴ Consistent with these findings, National Health and Nutrition Examination Survey data have demonstrated that adverse social determinants of health are associated with dyslipidemia.⁹⁸ This illustrates that for many people, hypertriglyceridemia is a manifestation of a complex set of factors, including dietary availability and choices, housing status, parental education, and health care access.

Therefore, comprehensive multilevel interventions, especially at the public policy level, are important to mitigate hypertriglyceridemia. Although altering lifestyle quality is a key medical goal, complementary goals may

include examining economic obstacles to obtaining healthy foods, engaging in parental education regarding cardiovascular risk factors, addressing school food policies on which youths may be dependent for adequate daily calories, advocating policy around safety and facilities for physical exercise, highlighting the dangers of environmental degradation, and influencing governmental policy around medical payers and access.^{99,100} More data are needed on the effects of societal interventions on ASCVD risk factors, especially their onset in childhood.

GAPS IN KNOWLEDGE AND CRITICAL AREAS FOR NEW RESEARCH/FUTURE ADVANCES

Current management of hypertriglyceridemia in youths relies on small pediatric studies and extrapolation from adult studies. Youth-focused studies are needed to determine which lifestyle measures are most efficacious for triglyceride reduction. These could include comparative efficacy of types of exercise (eg, aerobic versus resistance training) or specific dietary patterns (eg, Mediterranean versus low-carbohydrate diets). These findings should then be used to inform public policy in order to improve pediatric cardiometabolic health.

Triglyceride-lowering pharmacotherapies have not been well studied in youths. The most commonly used medications, that is, omega-3 FA and fibric acid derivatives, rely on small clinical trials^{77,78,101} or case series and have produced mixed results.^{102,103} Adequately powered double-blinded, randomized trials are needed to better assess the efficacy and safety of these medications in youths with hypertriglyceridemia. The role of statin medications in the treatment of elevated triglyceride and elevated non-HDL-C in youths requires further study. Furthermore, medications currently being evaluated in randomized trials for adult patients with hypertriglyceridemia require similar trials in youths.

CONCLUSIONS

Hypertriglyceridemia is frequently found in childhood and is linked to the development of ASCVD and acute pancreatitis. Although pediatric lipid screening guidelines have traditionally focused on familial hypercholesterolemia, hypertriglyceridemia in children is increasingly recognized because it is concomitant with obesity and metabolic syndrome. Diagnosis and treatment of hypertriglyceridemia in youths depend on the expected predominantly circulating lipoprotein. Early detection and implementation of triglyceride-lowering lifestyle strategies is key for those with mild and moderate hypertriglyceridemia, with pharmacotherapy rarely indicated. Youths with severely or very severely elevated triglycerides should be evaluated by a pediatric lipid specialist

because they may benefit from pharmacotherapy, significant dietary restrictions, or both. Further studies are needed to determine which lifestyle measures are most efficacious in lowering triglycerides, and adequately powered, double-blinded trials are needed to assess the safety and efficacy of triglyceride-lowering pharmacotherapy in children with hypertriglyceridemia.

ARTICLE INFORMATION

The American Heart Association makes every effort to avoid any actual or potential conflicts of interest that may arise as a result of an outside relationship or a personal, professional, or business interest of a member of the writing panel. Specifically, all members of the writing group are required to complete and submit a Disclosure Questionnaire showing all such relationships that might be perceived as real or potential conflicts of interest.

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Disclosures

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Writing group member	Employment	Research grant	Other research support	Speakers' bureau/honoraria	Expert witness	Ownership interest	Consultant/advisory board	Other
Jonathan P. Wong	University of Toronto (Canada)	None	None	None	None	None	None	None
Amy L. Peterson	University of Wisconsin School of Medicine and Public Health	Regeneron Pharmaceuticals (PI in investigator-initiated research study)*	None	None	None	None	Novartis (unpaid)*; Mirum Pharmaceuticals (unpaid)*	None
Ambika P. Ashraf	University of Alabama at Birmingham	Aegerion Pharmaceuticals, Inc (site PI, MEASURE: Metreleptin Effectiveness and Safety Registry)†; NIH (co-PI for U01DK134966, Youth Onset T2D Consortium, Discovery study)†	None	None	None	None	Chiesi USA, Inc*	None
Jennifer Bachman	Boston Children's Hospital	None	None	None	None	None	None	None
Robert A. Hegele	Western University Roberts Research Institute (Canada)	None	None	None	None	None	Arrowhead*; Ionis*; HLS Therapeutics*	None
Maya Rashad	Cook Children's Health Care System	None	None	None	None	None	None	None
Andrew M. South	Wake Forest University School of Medicine	NIH (K23)†; NIH (R25)†; NIH (R01)*; NIH (R01)*	None	None	None	None	IDORSIA Pharmaceuticals US, Inc†; Kyowa Kirin Co, Ltd†; Trave Therapeutics, Inc†; Techspert.IO*; MJH Life Sciences*	Wake Forest University Health Sciences (associate professor)†
Andrew H. Tran	Nationwide Children's Hospital The Heart Center	None	None	None	None	None	None	None
Justin Zachariah	Baylor College of Medicine Texas Children's Hospital	NHLBI (PI)†	None	None	None	None	None	None

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*Modest.

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Reviewer	Employment	Research grant	Other research support	Speakers' bureau/honoraria	Expert witness	Ownership interest	Consultant/advisory board	Other
Julie A. Brothers	The Children's Hospital of Philadelphia	None	None	None	None	None	None	NIH/NHLBI (DSMB committee member for Orlistat clinical trial)*
Jacob Hartz	Levine Children's Hospital	None	None	None	None	None	None	None
Michael Khoury	University of Alberta (Canada)	None	None	None	None	None	Ultragenyx Pharmaceutical Inc*	None
Geetha Raghuvier	Children's Mercy Hospitals and Clinics	None	None	None	None	None	None	None
Gerald F. Watts	University of Western Australia (Australia)	Arrowhead (investigator)*	None	CSL Seeger*; Amgen*	None	None	Arrowhead*	None

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*Modest.

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